Polar Coordinates and Conversions

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting points**

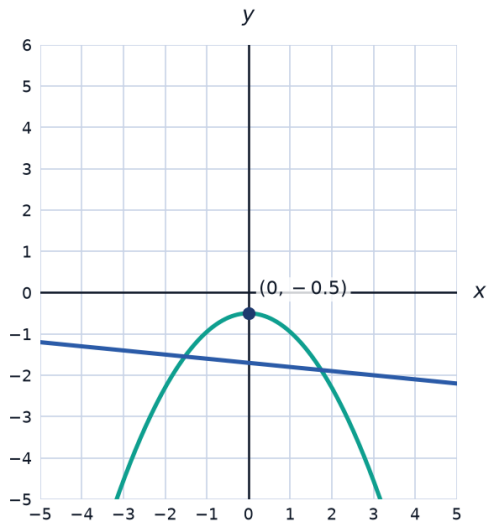
Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$

b $\left(2, \frac{3\pi}{4}\right)$

2 **[Foundational]** • **Target: Converting points**

Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:



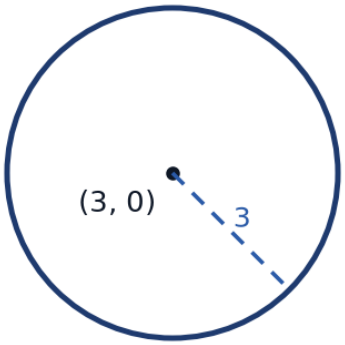
a $(-3, 3)$

b $(0, -5)$

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3 [Foundational] • Target: Converting equations

Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.



4 [Foundational] • Target: Converting equations

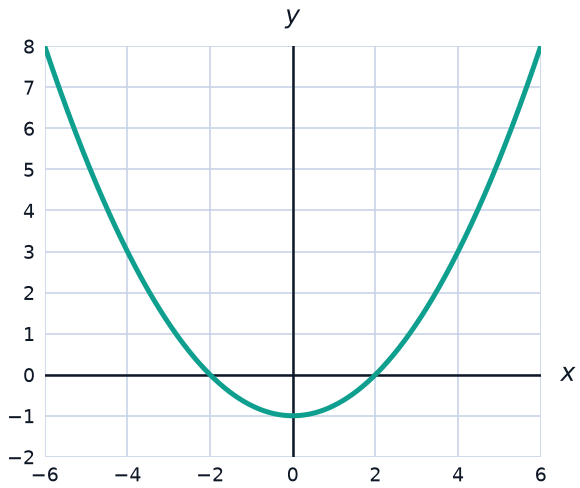
Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$

b $y = x$

5 [Foundational] • Target: Converting equations

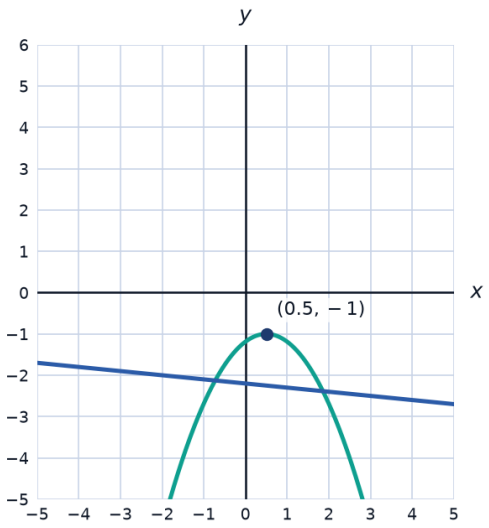
Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.



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6 [Foundational] • Target: Multiple representations

Give two other pairs (r, θ) that represent the same point as $(3, \frac{\pi}{4})$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.



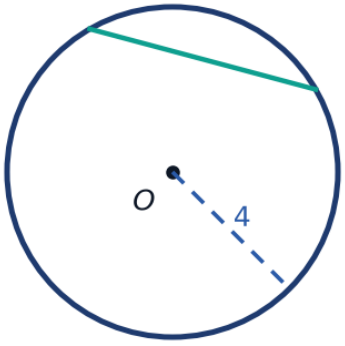
7 [Foundational] • Target: Multiple representations

Plot-reasoning: explain why the polar point $(-2, \frac{\pi}{2})$ corresponds to the same location as $(2, \frac{3\pi}{2})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distance between polar points

Convert $(4, \frac{\pi}{6})$ and $(4, \frac{2\pi}{3})$ to rectangular coordinates, then find the distance between them, to two decimal places.



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9 [Intermediate] • Target: Further conversion practice

Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.

10 [Intermediate] • Target: Why polar coordinates are not unique

Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.

11 [Intermediate] • Target: Why polar coordinates are not unique

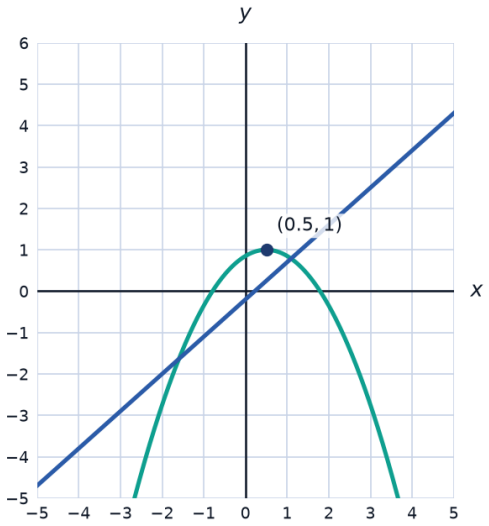
The pole (origin) is a special case in polar coordinates. Explain why.

12 [Intermediate] • Target: Converting more complex equations

Convert the rectangular equation $x^2 - y^2 = 4$ (a hyperbola) to polar form.

13 [Intermediate] • Target: More point conversions with negative r

Convert the polar point $\left(-4, \frac{\pi}{6}\right)$ (negative r) to rectangular coordinates.



14 [Intermediate] • Target: More point conversions with negative r

Convert the rectangular point $(-4, -4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

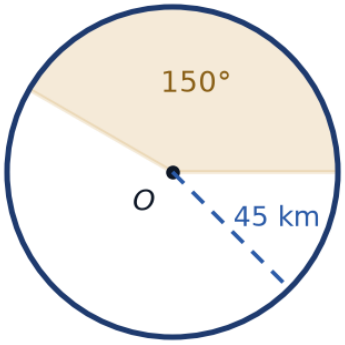
15 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 2.01 Right-Triangle Trigonometry

The Archimedean spiral $r = \theta$ has no simple rectangular equivalent, but explain qualitatively why it spirals outward as θ increases.

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16 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.02 Radian Measure and Arc Length

A radar station models detected objects using polar coordinates (r, θ) where r is distance in km and θ is bearing angle. An aircraft is detected at $(r, \theta) = (45, \frac{5\pi}{6})$. Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.



17 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

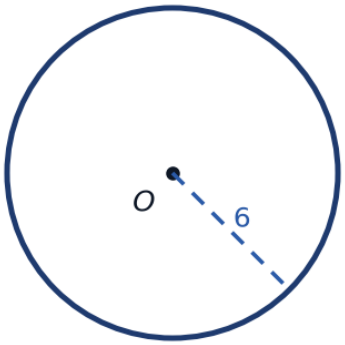
A second aircraft is detected at rectangular coordinates $(30, -40)$ relative to the same radar station. Convert to polar coordinates (r, θ) with $r > 0$ and $0 \le \theta < 2\pi$, to two decimal places.

18 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Convert the rectangular equation of a vertical line, $x = 5$, to polar form, and explain why the result is not simply a constant r.

19 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Convert the polar equation $r = 6$ to rectangular form, and explain in one sentence why this equation contains no θ term.



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20 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Level 2 · MediumFind all real solutions to the trigonometric equation $2\cos^2\theta + 3\sin\theta - 3 = 0$ on the interval $[0, 2\pi)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Converting points

The complex number z satisfies the equation $z(2 - \mathrm{i}) = 7 + 4\mathrm{i}$. Express z in the form $x + \mathrm{i}y$, where x and y are real.

a. $2 + 3\mathrm{i}$

b. $\frac{18}{5} + \frac{1}{5}\mathrm{i}$

c. $2 - 3\mathrm{i}$

d. $\frac{10}{3} + 5\mathrm{i}$

22 [Foundational] • Target: Converting equations

The function f is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?

a. 1 and 2

b. 0 and 1

c. 2 and 3

d. -1 and 0

23 [Foundational] • Target: Multiple representations

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A?

a. (-1, 0)

b. (5, 4)

c. (2, 0)

d. (0, -4)

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24 [Foundational] • Target: Distance between polar points

The unique real root α of the equation $x^3 + 2x - k = 0$, where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$

b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

25 [Foundational] • Target: Further conversion practice

The complex number z is given by $z = \frac{4 + 3\mathrm{i}}{2 - \mathrm{i}}$. Express z in the Cartesian form $x + \mathrm{i}y$, where x and y are real numbers.

a. $1 + 2\mathrm{i}$

b. $1 - 2\mathrm{i}$

c. $\frac{11}{5} + \frac{2}{5}\mathrm{i}$

d. $5 + 10\mathrm{i}$

26 [Foundational] • Target: Why polar coordinates are not unique

The points A, B, and C have coordinates A(2, 5), B(k, 1), and C(8, -3). If A, B, and C are collinear, find the value of k.

a. 4

b. 5

c. 6

d. 7

27 [Foundational] • Target: Converting more complex equations

For all real values of x such that $x \neq -1$ and $x \neq 2$, the equation $\frac{ax + 7}{x + 1} - \frac{b}{x - 2} = \frac{2x^2 - 3x - 20}{x^2 - x - 2}$ is true, where a and b are constants. What is the value of $a + b$?

a. 4

b. 6

c. 8

d. 10

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28 [Foundational] • Target: More point conversions with negative r

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19

b. 31

c. 16

d. 11

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Converting a spiral equation

Consider the quadratic equation $2x^2 - (k-1)x + (k+1) = 0$ where k is a real constant. The equation has two non-real complex conjugate roots. If the imaginary parts of the roots have a magnitude of $\frac{\sqrt{7}}{2}$, what is the greatest possible value of k?

a. 7

b. 3

c. 10

d. 5

30 [Intermediate] • Target: Converting a real-world radar equation

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

a. $x = 3 \pm 2i$

b. $x = 3 \pm 4i$

c. $x = -3 \pm 2i$

d. $x = 3 \pm 2$

31 [Intermediate] • Target: Converting points

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?

a. $x = 3$ and $x = 5$

b. $x = -3$ and $x = -5$

c. $x = 3$ and $x = -5$

d. $x = -3$ and $x = 5$

32 [Intermediate] • Target: Converting equations

Which is NOT a quadratic equation?

a. $2x^2 = 5$

b. $x^2 + x + 1 = 0$

c. $x^2 - 1 = 0$

d. $x^3 + 2x = 0$

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33 [Intermediate] • Target: Multiple representations

An ellipse has equation $\frac{x^2}{36} + \frac{y^2}{20} = 1$. What are the coordinates of its foci?

a. $(\pm 5, 0)$

b. $(\pm 4, 0)$

c. $(0, \pm 4)$

d. $(\pm 6, 0)$

34 [Intermediate] • Target: Distance between polar points

Find the x-coordinates of the points of intersection of $y = x^2 - 9x + 22$ and $y = 2$.

a. x=4 or x=5

b. x=-4 or x=-5

c. x=5 or x=5

d. x=4 or x=4

35 [Intermediate] • Target: Further conversion practice

The line $y = -12$ intersects the circle $x^2 + y^2 = 169$ at two points. What is the distance between the two intersection points?

a. 10

b. 13

c. 16

d. 5

36 [Intermediate] • Target: Why polar coordinates are not unique

If z and $\bar{z}$ are roots of $x^2 + bx + c = 0$ (real coefficients), express b and c in terms of $a + bi$.

a. $b = 0,\ c = a^2$

b. $b = a,\ c = b$

c. $b = -2a,\ c = a^2 + b^2$

d. $b = 2a,\ c = a^2 - b^2$

37 [Intermediate] • Target: Converting more complex equations

Which equation has roots that are complex conjugates?

a. $x^2 - 5 = 0$

b. $x^2 + 2x - 5 = 0$

c. $x^2 - 2x + 5 = 0$

d. $x^2 - 2x - 5 = 0$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: More point conversions with negative r • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the distance between the points $(7, 4)$ and $(16, -8)$.

a. 21

b. 14

c. 15

d. 16

39 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Derive the polar equation of a conic with focus at the pole and directrix perpendicular to the polar axis at distance d.

a. $r = d/\theta$

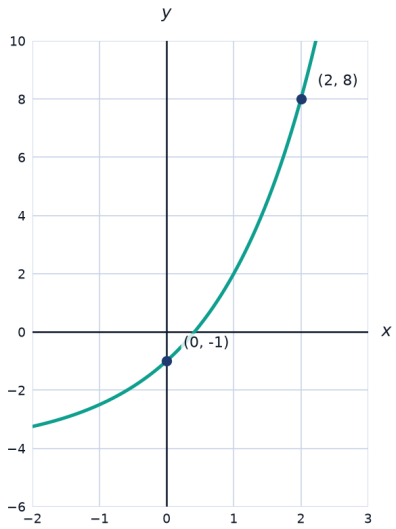
b. $r = e/(1 - d\cos\theta)$

c. $r = ed\cos\theta$

d. $r = \frac{ed}{1 - e\cos\theta}$ (directrix to the left)

40 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?



a. $-\frac{5}{2}$

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

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41 [Challenging] • Target: Converting points • Cross-Lesson: 3.03 Sum and Difference Formulas

Consider the matrix equation $(2, 1, -2; 4, -1, -1; 2, -2, 1)(x; y; z) = (9; 9; k)$ where k is a real constant. Given that the system has infinitely many real solutions (x, y, z), what is the minimum possible value of $x^2 + y^2 + z^2$?

a. 9

b. 3

c. 6

d. 45

42 [Challenging] • Target: Converting equations • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Consider the function $f(x) = 4\cos^2(x) + b\sin(x) + c$ where b and c are real constants with $b > 0$. If the maximum value of $f(x)$ is 10 and the minimum value of $f(x)$ is 1, what is the value of c?

a. 5

b. $\frac{11}{2}$

c. 9

d. 4

43 [Challenging] • Target: Multiple representations • Cross-Lesson: 4.01 Circles and Parabolas

Determine the complete set of real values of the constant k for which the equation $2\cos 2\theta - 4\sin\theta = k$ has at least one real solution for θ.

a. $-6 \le k \le 3$

b. $-6 \le k \le 2$

c. $k \le 3$

d. $-3 \le k \le 6$

44 [Challenging] • Target: Distance between polar points • Cross-Lesson: 4.02 Ellipses

Find the set of all real solutions to the equation $\log_2(x - 1) - \log_4(x + 14) = 1$

a. $x = 11$ only

b. $x = 11$ and $x = -5$

c. $x = 5$ only

d. $x = 16$ only

45 [Challenging] • Target: Further conversion practice • Cross-Lesson: 4.03 Hyperbolas

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3

b. 1

c. 3

d. 24

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